

CO3-AT1 – DATA HANDLING and Visualization

DESIGN TASK

1. A healthcare organization has collected patient waiting-time data from 20 hospitals over one year.

Task: Design a visualization dashboard that:

- Shows Box Plots and Violin Plots for waiting times across hospitals.
- Identifies outliers and extreme waiting-time cases.
- Highlights differences in median waiting times and variability among hospitals.

Objective: Help hospital administrators identify service inefficiencies and improve patient care quality.

ANSWER:

```
set.seed(123)

hospital <- rep(paste("Hospital", 1:20), each = 50)
waiting_time <- rnorm(1000, mean = 30, sd = 8)
data <- data.frame(hospital, waiting_time)

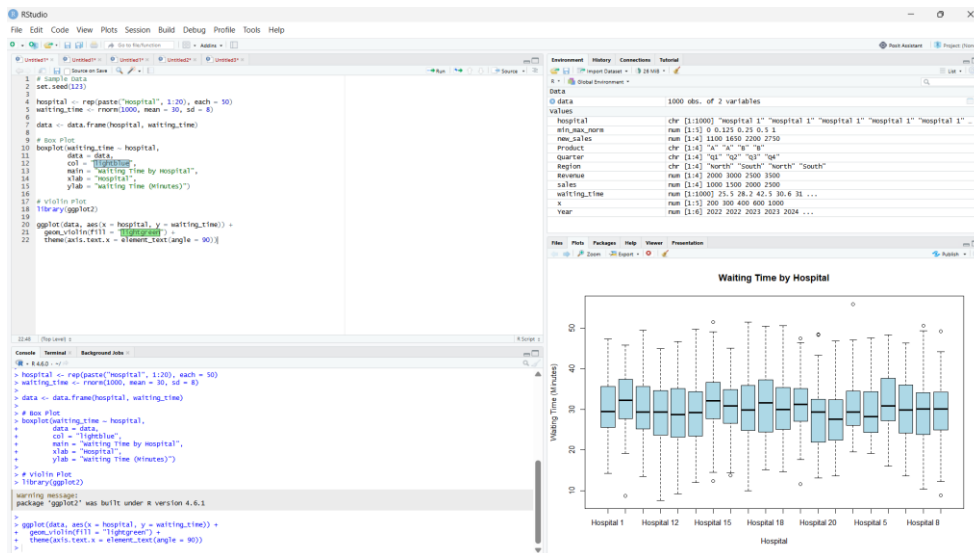
boxplot(waiting_time ~ hospital,
        data = data,
        col = "lightblue",
        main = "Waiting Time by Hospital",
        xlab = "Hospital",
        ylab = "Waiting Time (Minutes)")

library(ggplot2)

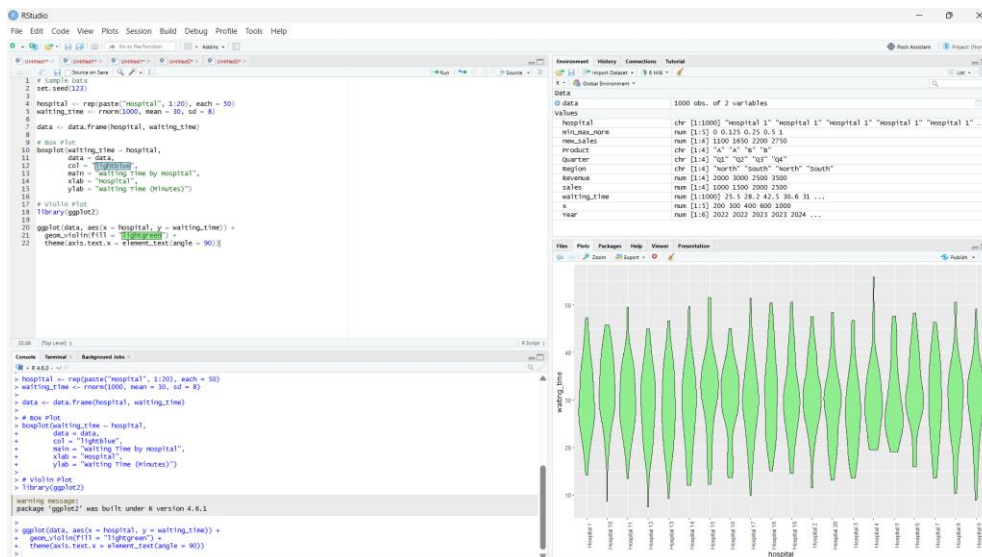
ggplot(data, aes(x = hospital, y = waiting_time)) +
  geom_violin(fill = "lightgreen") +
  theme(axis.text.x = element_text(angle = 90))
```

OUTPUT:

Box Plot:



Violin Plot:



2. A smart city project monitors hourly air quality data (PM2.5, PM10, NO₂, and CO) from 15 monitoring stations.

Task: Create a visualization design that:

- Uses Heatmaps to display pollution intensity across locations and time periods.
- Employs Ridgeline Plots to compare pollutant distributions among stations.
- Highlights pollution spikes and abnormal environmental conditions.

Objective: Assist environmental authorities in identifying pollution hotspots and planning mitigation measures.

ANSWER:

```
library(ggplot2)
```

```
station <- rep(paste("S",1:15), each=24)
```

```
hour <- rep(1:24,15)
```

```
pm25 <- sample(20:150,360,replace=TRUE)
```

```
air <- data.frame(station,hour,pm25)
```

```
ggplot(air, aes(hour, station, fill = pm25)) +
```

```
  geom_tile() +
```

```
  labs(title="PM2.5 Heatmap")
```

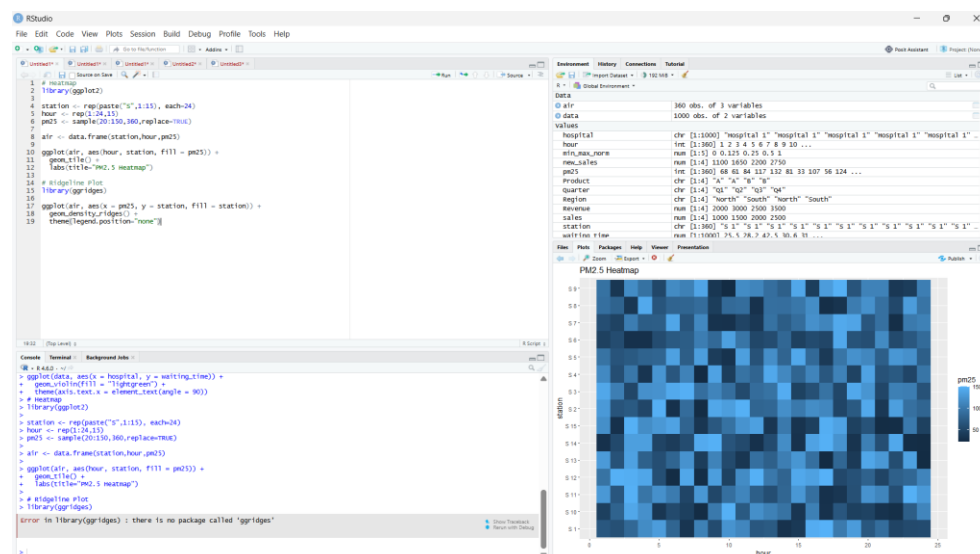
```
library(ggribes)
```

```
ggplot(air, aes(x = pm25, y = station, fill = station)) +
```

```
  geom_density_ridges() +
```

```
  theme(legend.position="none")
```

OUTPUT:



3. A university tracks student academic performance across multiple departments over five academic years.

Task: Design a visualization dashboard that:

- Uses Line Charts to show performance trends over time.
- Incorporates Scatter Plots to analyze the relationship between attendance and grades.
- Uses Facet Grids to compare departmental performance simultaneously.

Objective: Enable academic administrators to identify performance trends and factors affecting student success.

ANSWER:

```
library(ggplot2)
```

```
year <- rep(2020:2024,4)
```

```
department <- rep(c("CSE","ECE","IT","MECH"), each=5)
```

```
gpa <- c(7.2,7.4,7.6,7.8,8.0,
```

```
        6.9,7.1,7.3,7.5,7.6,
```

```
        7.0,7.2,7.5,7.7,7.9,
```

```
        6.8,7.0,7.1,7.2,7.3)
```

```
attendance <- sample(60:100,20)
```

```
grade <- sample(60:100,20)
```

```
student <- data.frame(year,department,gpa,attendance,grade)
```

```
ggplot(student, aes(year, gpa, color=department)) +
```

```
  geom_line() +
```

```
  geom_point()
```

```
ggplot(student, aes(attendance, grade)) +
```

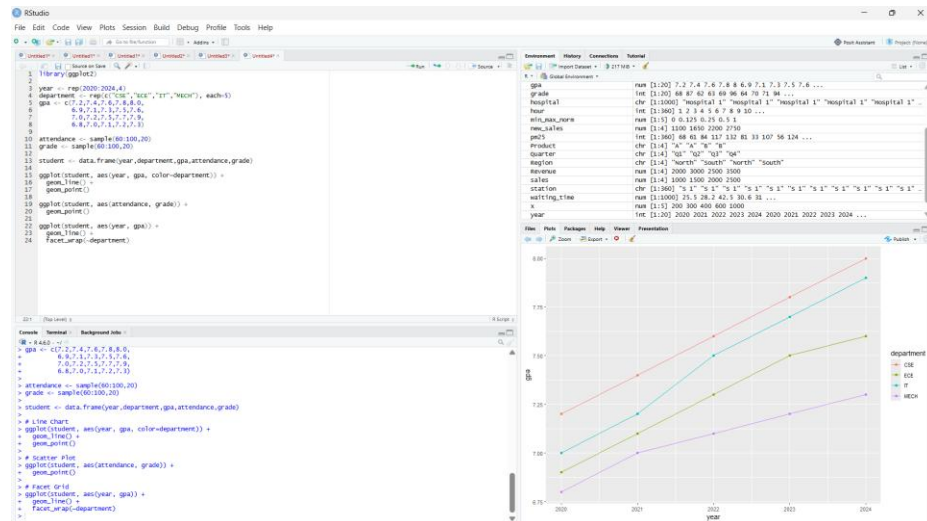
```
  geom_point()
```

```
ggplot(student, aes(year, gpa)) +
```

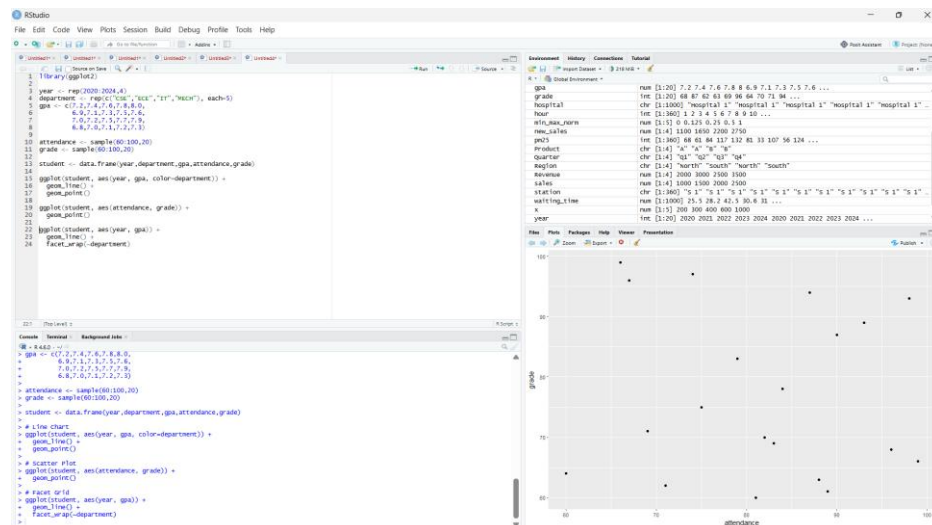
```
  geom_line() +
```

```
  facet_wrap(~department)
```

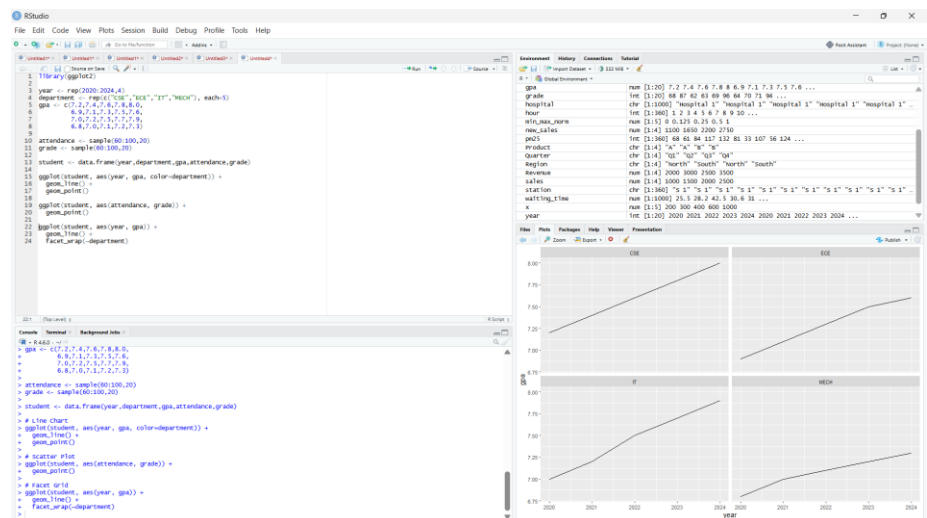
OUTPUT: Line Plot:



Scatter Plot:



Facet Grid:



4. A logistics company records delivery times for thousands of shipments using Road, Rail, Air, and Sea transportation modes.

Task: Create a visualization design that:

- Uses Histograms and ECDFs to compare delivery-time distributions across transportation modes.
- Employs Q-Q Plots to assess whether delivery times follow a theoretical normal distribution.
- Highlights delays, variability, and outliers in delivery performance.

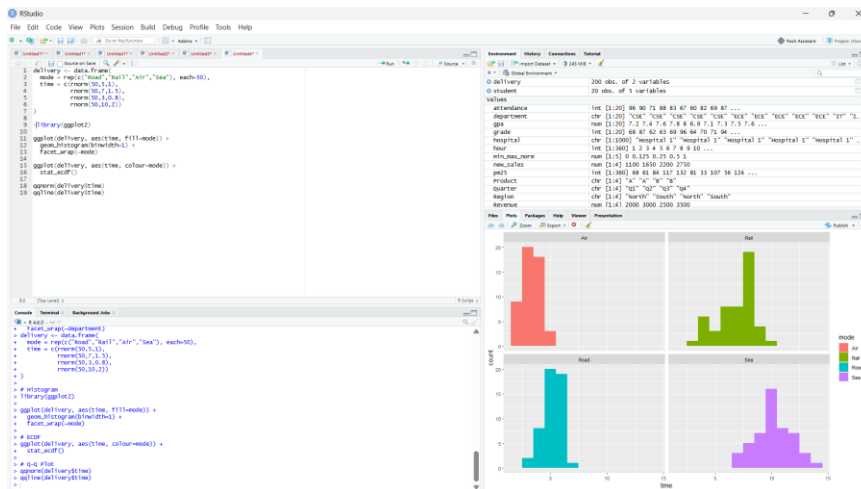
Objective: Support operational teams in optimizing transportation strategies and improving delivery reliability.

ANSWER:

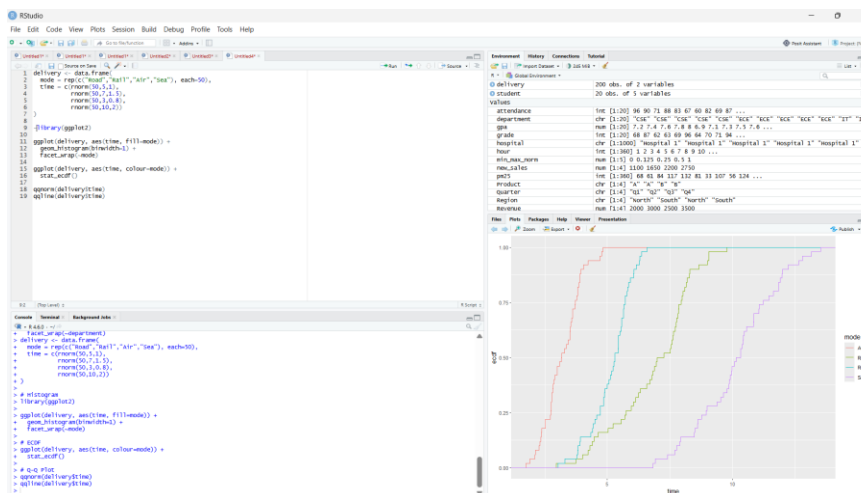
```
delivery <- data.frame(  
  mode = rep(c("Road", "Rail", "Air", "Sea"), each=50),  
  time = c(rnorm(50,5,1),  
           rnorm(50,7,1.5),  
           rnorm(50,3,0.8),  
           rnorm(50,10,2))  
)  
  
library(ggplot2)  
  
ggplot(delivery, aes(time, fill=mode)) +  
  geom_histogram(binwidth=1) +  
  facet_wrap(~mode)  
  
ggplot(delivery, aes(time, colour=mode)) +  
  stat_ecdf()  
  
qqnorm(delivery$time)  
qqline(delivery$time)
```

OUTPUT:

Histogram:



ECDF:



Q-Q Plot:

